

CONDITIONAL POPULATION BRIDGE

Common radius: coordinates, overlaps, and cancellation

Use the full hypotheses and density expansions in the adjacent text.

STEP 1 · COORDINATES

Keep the sources distinct

Separate source coordinates
in the same measured space M .

Source 1 coordinate



radius r
ball volume v_r

Source 2 coordinate



radius r
ball volume v_r

Target coordinate in N

Same radius r .
Ball volume w_r .

STEP 2 · OVERLAPS

Control both event overlaps

Source overlap

$$\Pr(E_{1,r} \cap E_{2,r}) \\ = o(v_r)$$

Target-source triple overlap

$$\Pr(C_r \cap E_{1,r} \cap E_{2,r}) \\ = o(w_r v_r)$$

Local essential bounds
are sufficient conditions.
They are not necessary.

STEP 3 · CANCELLATION

Cancel the common factor

Common volume factor

$$w_r v_r$$

Use inclusion-exclusion.
Numerator and denominator
share the factor $w_r v_r$.

Cancel it in the ratio.
Then take the natural log.

Positive limiting density
sums are required.

FIRST-ORDER OVERLAP · FAILURE BOUNDARY

If $\Pr(E_{1,r} \cap E_{2,r}) = \kappa v_r + o(v_r)$, with $\kappa > 0$, the union denominator changes.

The stated density ratio then need not follow. The full hypotheses and proof are in the text.

Conditional population lemma; no kNN or finite-sample guarantee.